TIER-ITS

Theory-Indexed Evidence Retrieval for Long Mental-Health Conversations

Can an ITS-based memory help an AI remember important risk evidence?

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Proposal in one paragraph. People do not always describe distress in one clear sentence. Important information may appear early in a long conversation and later be surrounded by unrelated details. Recent research shows that large language models (LLMs) become worse at detecting suicidal ideation as a conversation grows and changes direction [7]. This project will test a simple solution: keep a small, structured memory of evidence related to the Interpersonal Theory of Suicide (ITS). The memory will record what was said, when it was said, whether it is current or historical, and whether it was later denied or corrected. Retrieval-augmented generation (RAG) will bring the relevant evidence back when the model prepares a report. The project is a research prototype tested on existing data and controlled simulated conversations. It will not talk to live users, diagnose anyone, or make a real clinical decision.

1 The Problem

An LLM can read a short message and assign a risk label. A long conversation is harder. A person may mention feeling like a burden near the beginning, discuss unrelated events for many turns, and later describe being alone. A model reading the whole conversation may overlook the early statement. A model reading only the latest turns will not see it at all.

Kalinich et al. placed clinician-checked statements at different positions inside long psychotherapy transcripts. Across 49 LLMs, detection became worse as the relevant statement appeared deeper in the conversation. The direction and content of the surrounding conversation also mattered, not only its length [7]. This gives the project a clear problem to study:

> **Can a small, theory-based memory preserve important evidence better than giving an LLM the full conversation or a general summary?**

The proposed memory is based on the Interpersonal Theory of Suicide. ITS describes several states that may contribute to suicidal desire or dangerous behaviour: thwarted belongingness (feeling disconnected), perceived burdensomeness (feeling like a burden), hopelessness about these states, and acquired capability related to fearlessness about death and tolerance of pain [16]. In this proposal, these states are treated as research concepts. Detecting them in text is only an approximation of validated self-report measures.

ITS will help organise and explain evidence. It will *not* set a clinical risk level by itself. A structured instrument such as the Columbia Suicide Severity Rating Scale (C-SSRS) asks about suicidal thoughts and behaviour [12], while a framework such as SAFE-T combines inquiry with risk factors, protective factors and professional judgement [13]. The prototype will report evidence for a human reviewer; it will not replace that judgement.

2 Review of Existing Work

2.1 Text classifiers

Most earlier systems read a post or a group of posts and predict a label. Examples include the CLPsych shared tasks and domain-specific language models such as MentalBERT [17, 6]. These systems are useful baselines, but they normally do not maintain a changing memory across a conversation.

Gaur et al. created a dataset of Reddit users labelled with C-SSRS-based severity levels [4]. This is more useful for evaluation than datasets whose labels are taken only from the subreddit in which a post appeared. However, it is still post-based rather than a real intake conversation.

2.2 ITS-based text analysis

Garg et al. released 3,522 posts with human labels and explanations for thwarted belongingness and perceived burdensomeness [3]. InterPrompt used LLM prompts to identify these related factors [14]. Shimgekar et al. studied ITS concepts in a much larger collection of SuicideWatch posts [15]. Therefore, extracting ITS concepts from text is not new.

The open question is different: can these concepts be used as a simple memory that remains useful across a long and changing conversation?

2.3 LLM agents for mental-health assessment

Recent systems already use several LLM agents. AgentMental separates tasks such as asking questions, checking whether enough information has been collected, scoring answers and updating memory [5]. Other work has studied adaptive multi-agent screening [9] and structured psychiatric interviews with several cooperating agents [8]. Therefore, using multiple agents is also not new.

This proposal does not claim that an agent, a memory or RAG is new. It asks whether a particular kind of memory—an evidence-linked ITS memory—solves a measured long-conversation problem.

2.4 Clear novelty claim

The proposed contribution has three parts:

1. A small **ITS evidence ledger** that records the concept, the exact supporting turn, time, subject, uncertainty and later contradiction.
2. A controlled test of whether this ledger reduces the loss of risk evidence as conversations become longer.
3. A faithfulness test: when a concept in the ledger is corrected or removed, the final report should change only when the evidence justifies a change.

The claim is deliberately narrow. The project does not claim to invent ITS extraction, RAG, agentic assessment or suicide-risk classification. It tests whether a theory-indexed memory is more reliable and easier to inspect than common alternatives.

Table 1. How the proposed study differs from nearby work.

Work	Main purpose	What it already provides	Gap used here
ITS datasets and prompts [3, 14, 15]	Find ITS concepts in posts	Construct labels and explanations	Do not test memory across long conversations
AgentMental [5]	Adaptive mental-health questioning	Multiple agents and a tree-like memory	Not an ITS-based suicide-risk memory study
Structured multi-agent interviewing [8]	Run organised psychiatric screening	Protocol-based agents and audit records	Does not test the loss of early evidence using ITS concepts
Long-conversation benchmark [7]	Measure whether detection gets worse with depth	Controlled evidence placed at different depths	Measures the failure but does not test an ITS memory as the remedy
TIER-ITS	Preserve and retrieve theory-linked evidence	ITS ledger, conversation RAG and concept correction	Proposed work

3 Aim, Questions and Expected Results

Aim. To test whether an evidence-linked ITS memory helps an LLM preserve, retrieve and correctly use important information in long mental-health conversations.

3.1 Research questions

- **RQ1: Memory.** Does the ITS ledger reduce the drop in performance when evidence appears early in a long conversation?
- **RQ2: Comparison.** Is it better than giving the model the full conversation, the latest turns, or a general summary?
- **RQ3: Evidence.** Can the system point to the correct conversation turn for each ITS concept in its report?
- **RQ4: Faithfulness.** If evidence is removed, corrected or changed, does the report change in the expected way?

3.2 Expected results

The main hypothesis is that the ITS ledger will lose less performance than the other methods as relevant evidence is moved deeper into a conversation. It should also retrieve the correct evidence more often than a general summary.

This is a test, not an assumption. A valid result may show that the ledger improves evidence retrieval but does not improve final classification. It may also show that some ITS concepts are useful while others cannot be inferred reliably from conversation text.

4 Proposed System

4.1 The ITS evidence ledger

After each conversation turn, an LLM extractor produces zero or more evidence records. Each record contains:

- the ITS concept, such as thwarted belongingness or perceived burdensomeness;

- whether the evidence supports, rejects or leaves the concept uncertain;
- the exact turn and text span that produced the record;
- whether the statement concerns the speaker, another person or a quotation;
- whether it concerns the present or the past;
- a confidence score; and
- a link to any later statement that corrects or contradicts it.

For example, "My brother says he is a burden" should not become evidence that the speaker feels like a burden. Likewise, "I used to feel completely alone, but I do not now" should be stored as historical evidence together with a current correction.

The ledger is a simple form of a *concept bottleneck*. Instead of moving directly from raw text to a final output, the system first records human-readable concepts. This makes it possible to inspect and correct the concepts before producing the report [10].

4.2 Two kinds of retrieval

The project uses RAG in two small and clear ways:

1. **Conversation retrieval.** When preparing a report, the system retrieves earlier turns linked to relevant ITS concepts. This is the main contribution.
2. **Protocol retrieval.** The system retrieves the relevant C-SSRS or SAFE-T passage so that the report can cite the basis for a question or field. RAG is used for provenance, not because the protocol documents are large [11].

4.3 Limited agent behaviour

An agent checks whether the ledger contains missing or contradictory information. In simulated conversations only, it may select one bounded follow-up question from a reviewed question bank. It cannot freely invent questions, provide advice, contact anyone or make a clinical decision.

This agent is a secondary feature. The main experiment concerns memory, so the project can still succeed if the questioning feature is not completed.

4.4 Output

The final output is a research report containing:

- the detected ITS concepts;
- the exact supporting and contradicting turns;
- missing or uncertain information;
- the retrieved protocol references;
- a model prediction used only for evaluation; and
- an explicit "insufficient evidence" result when confidence is low.

The report does not contain a hidden chain of thought. The audit record consists of observable items: input turns, ledger updates, retrieved passages, tool calls and final fields.

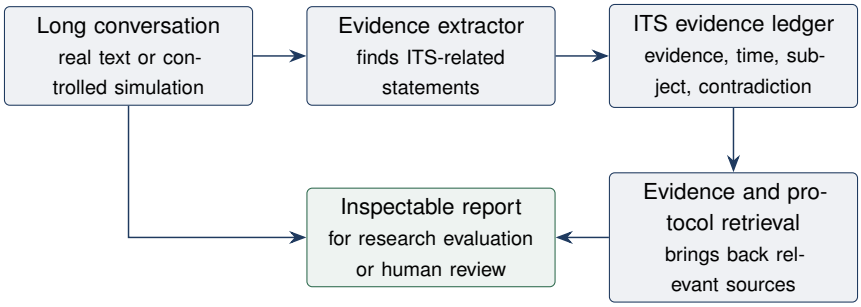


Figure 1. Simple architecture. The direct path from the conversation to the report is used as a baseline.

5 Data and Test Set

5.1 Existing datasets

The project will use different datasets for different questions. They will not be treated as interchangeable.

Table 2. Planned data sources and their limited roles.

Source	Use in this project	Important limitation
IRF dataset [3]	Test extraction of TB and PB	Post-level data; no acquired-capability or clinical risk label
C-SSRS-labelled Reddit data [4]	Secondary test of severity prediction	Small, retrospective and based on online posts
CLPsych evidence task [1]	Optional test of evidence-span retrieval	Restricted data access may delay use
Controlled long conversations	Main memory experiment	Synthetic construction does not show clinical effectiveness
Protocol documents [12, 13]	Small retrieval collection	A protocol citation does not replace professional judgement

Subreddit-derived binary labels may be used for software testing, but they will not support clinical claims. The IRF data will support claims about TB and PB, not claims about overall risk level.

5.2 Controlled long-conversation benchmark

The main test set will contain approximately 160 target statements, balanced between statements with and without relevant evidence. The statements will be placed at several conversation depths, for example 0, 20, 50 and 100 turns. Surrounding text will come from permitted retrospective or synthetic dialogue that has been checked not to add method details or an unintended second target.

The benchmark will include difficult but realistic language cases:

- relevant evidence followed by a long unrelated discussion;
- historical evidence versus a current feeling;
- a quotation or statement about another person;
- negation, uncertainty and sarcasm;

- an early statement that is later corrected;
- separate TB and PB evidence appearing far apart; and
- protective information introduced after concerning evidence.

The target label and target evidence span are known by construction. Two reviewers will independently check a sample of the completed conversations to ensure that the surrounding context has not changed the intended meaning. Any conversation with disagreement will be corrected or removed.

6 Experiments

6.1 Methods to compare

The same model will be used where possible so that the memory method, rather than model size, is the main difference.

1. **Target only:** the model sees only the target statement. This estimates the easiest possible condition.
2. **Full conversation:** the model receives all turns at once.
3. **Recent window:** the model sees only the latest fixed number of turns.
4. **General summary:** an LLM maintains a normal prose summary.
5. **TIER-ITS:** the model uses the structured ITS evidence ledger and retrieves the relevant turns.
6. **Gold ledger:** the ledger is filled with correct human labels. This shows the best result possible if extraction errors were removed.

The gold-ledger condition is important. If it performs well but the automatic ledger does not, the main problem is concept extraction. If even the gold ledger does not help, the proposed memory design itself is not useful.

6.2 Primary experiment

For each method, performance will be measured as the target statement is moved deeper into the conversation. The primary result is the change in F1 score from depth 0 to depth 100. Results will also be reported separately for positive and negative cases so that a high score cannot hide missed concerning evidence.

6.3 Evidence test

The report must name the correct turn or turns that support each concept. Evidence precision measures how often a cited turn is relevant. Evidence recall measures how much of the labelled evidence was found.

6.4 Correction and contradiction test

Pairs of conversations will differ in one controlled way. For example, one version may say “I still feel like a burden,” while the other says “I used to feel that way, but I do not now.” The test checks whether the ledger updates the time and status rather than simply keeping both statements as equally current.

6.5 Concept-intervention test

The value of one ledger concept will be corrected before the report is produced. The experiment asks whether the output changes in a reasonable and traceable way. This follows the central idea of concept-bottleneck models: a human-understandable concept can be inspected and edited before the final prediction [10].

6.6 Metrics and analysis

The main measures are:

- F1, recall and precision at each conversation depth;
- the drop in F1 between short and long conditions;
- evidence precision and evidence recall;
- accuracy on subject, time, negation and contradiction fields;
- confidence calibration, meaning whether high confidence usually corresponds to correct results;
- the rate of "insufficient evidence" outputs; and
- processing time and API cost.

Paired bootstrap confidence intervals will compare methods on the same cases [2]. The number of cases in every subgroup will be shown. This is an engineering and robustness study, not a clinical non-inferiority trial.

7 Feasibility

7.1 Why the project is manageable

The revised project has one main question and one main experiment. It does not require training a large language model, collecting live patient data or building a full clinical chatbot. The main components are structured LLM output, a small evidence store and simple retrieval.

The long-conversation benchmark is also controllable. Conversation depth can be changed without collecting new participants, and the same target cases can be used for every method. Response caching will limit cost and make repeated experiments reproducible.

7.2 Eight-week plan

Table 3. Practical work plan.

Weeks	Main work	Required output
1	Final literature review; data and ethics checks	Frozen research questions and dataset list
2	Build the controlled conversation generator and review rules	Small pilot benchmark
3	Implement the extractor and ITS ledger	Valid structured records with evidence links
4	Add conversation retrieval and protocol retrieval	Working TIER-ITS pipeline
5	Implement full-context, recent-window and summary baselines	All comparison methods frozen
6	Run the main depth and evidence experiments	Complete result files and error cases
7	Run correction and concept-intervention tests	Statistical analysis and figures
8	Write the paper; document limitations and reproducibility	Final paper and research artefacts

The follow-up-question agent is optional. It will be removed first if the schedule slips.

7.3 Main risks

Table 4. Main project risks and responses.

Risk	Response
The extractor confuses the speaker with another person	Store the subject explicitly and include quotation cases in the benchmark
The ledger keeps old evidence after a correction	Store time and contradiction links; test paired correction cases
Synthetic context is unrealistic	Report it as a controlled stress test, not as clinical validation
The surrounding text changes the target meaning	Human-check a sample; remove ambiguous cases
The project becomes too large	Keep memory as the core; drop active questioning and optional datasets
Restricted data are unavailable	Complete the main controlled benchmark and use public IRF data
The model produces unsafe or graphic text	Use fixed, reviewed prompts and exclude method or means details from generated material

8 Ethics and Safety

The system is not intended for direct use with people in crisis. It will be tested only on permitted retrospective text and controlled simulations. Any institutional ethics review or exemption decision will be obtained before annotation begins.

The following safeguards apply:

- No live participant will receive a model-generated risk decision.
- Examples in the paper will be paraphrased and stripped of identifying details.
- Generated conversations will avoid graphic method or means details.
- Reviewers may pause or stop because the material can be distressing.

- Raw user text will not be sent to an external API unless its terms and the dataset agreement allow it.
- The paper will not claim clinical effectiveness, diagnosis or deployment readiness.

The main ethical advantage of the proposed design is inspectability. Each concept must point back to evidence, uncertainty is visible, and "insufficient evidence" is an allowed result. These features do not make the system clinically safe, but they make research errors easier to find.

9 Expected Contribution

The project will produce:

1. a controlled benchmark for studying the loss of risk-related evidence in long conversations;
2. a simple ITS evidence-ledger design with time, subject, uncertainty and contradiction fields;
3. a comparison against full-context, recent-window and general-summary baselines;
4. tests showing whether explanations point to the correct evidence and respond to concept corrections; and
5. a clear account of when ITS concepts help, fail or cannot be inferred from text.

The desired paper is therefore not "we built another mental-health chatbot." Its contribution is a focused answer to a measurable question:

> **Does a theory-based, evidence-linked memory help an LLM remember the right information in a long mental-health conversation?**

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